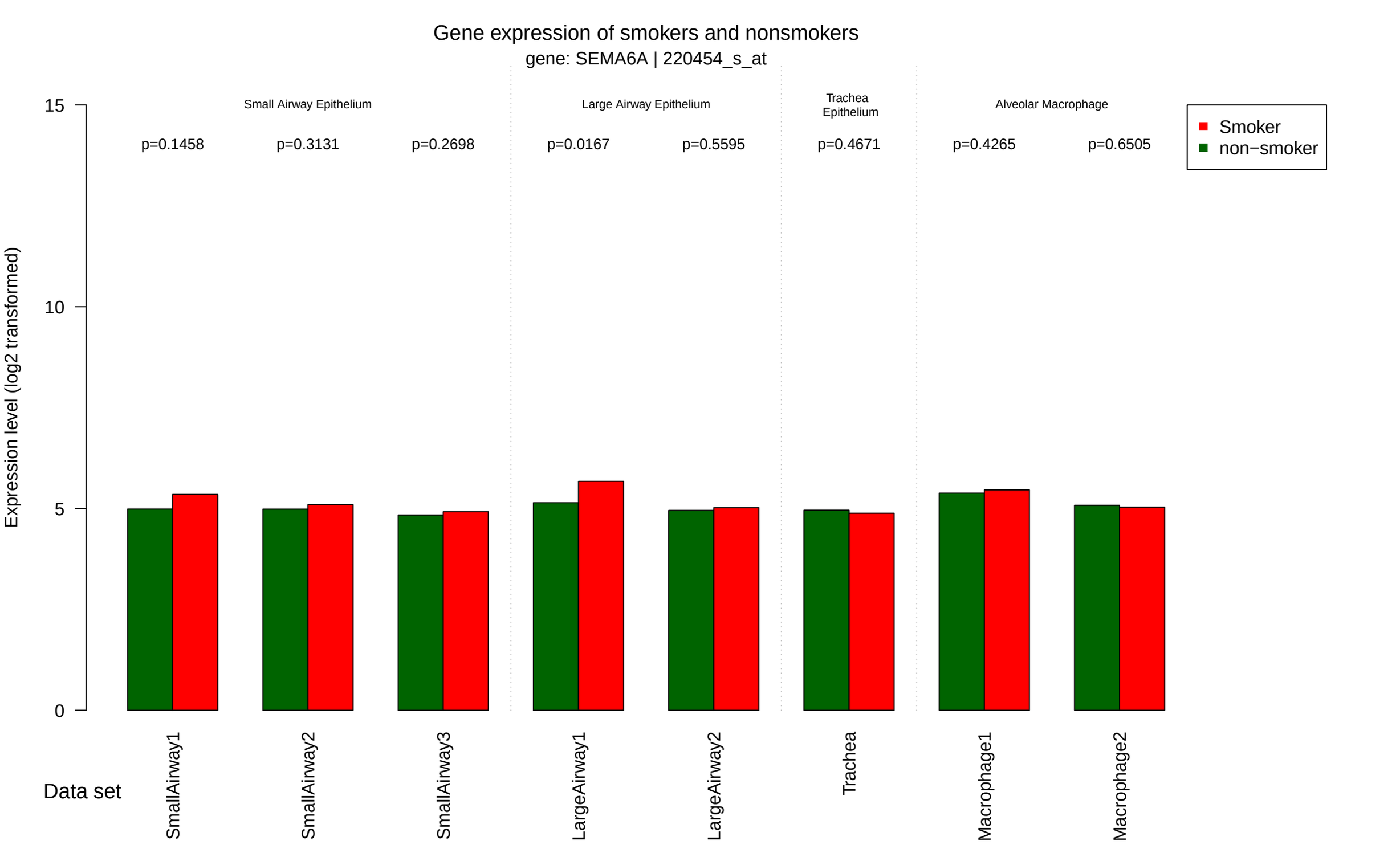
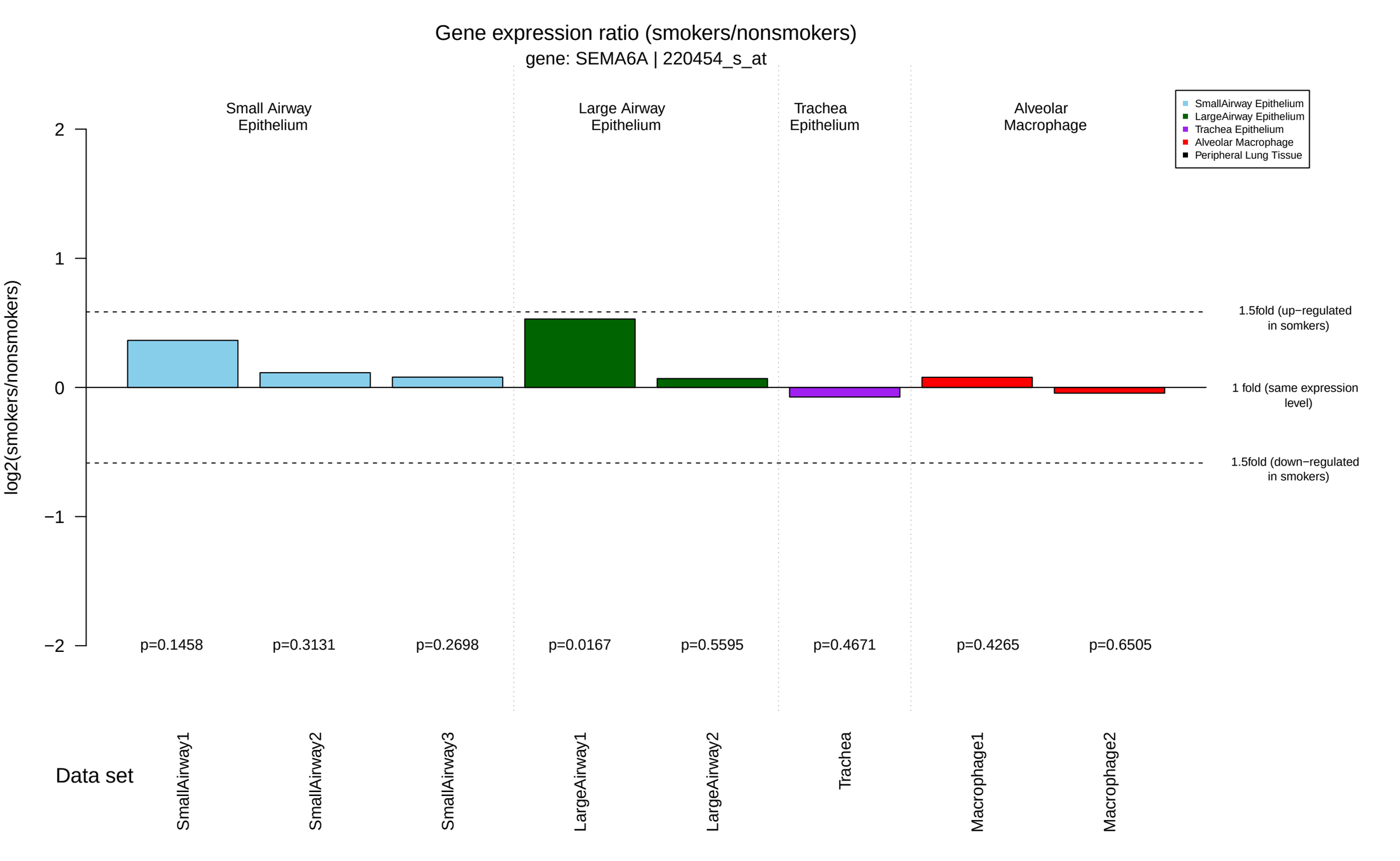
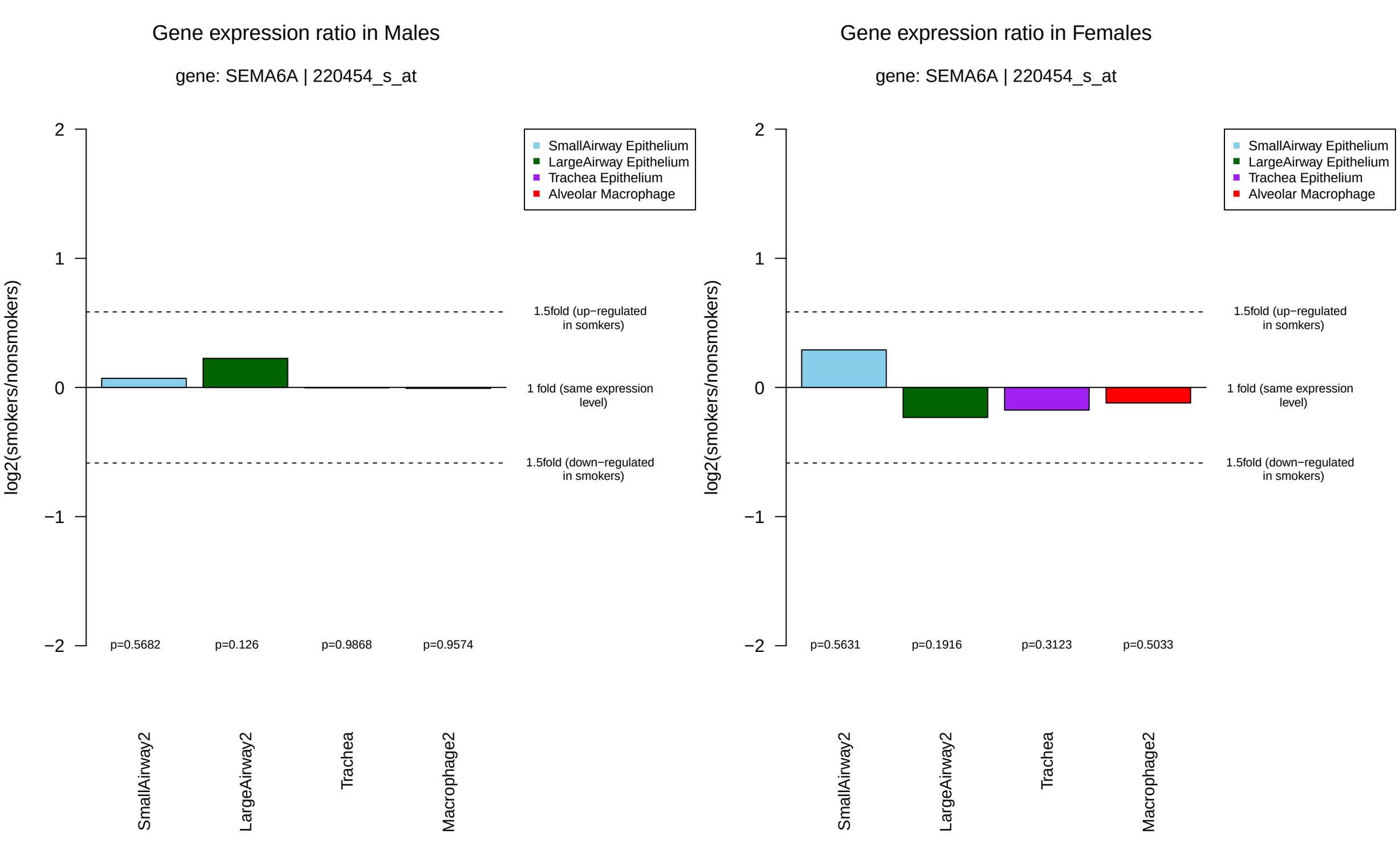
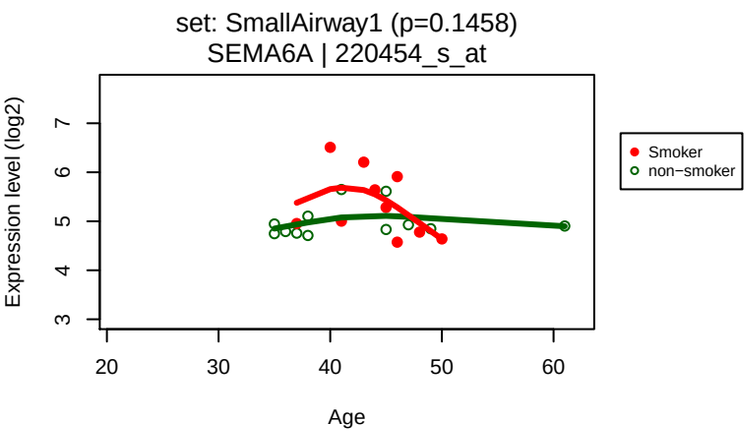
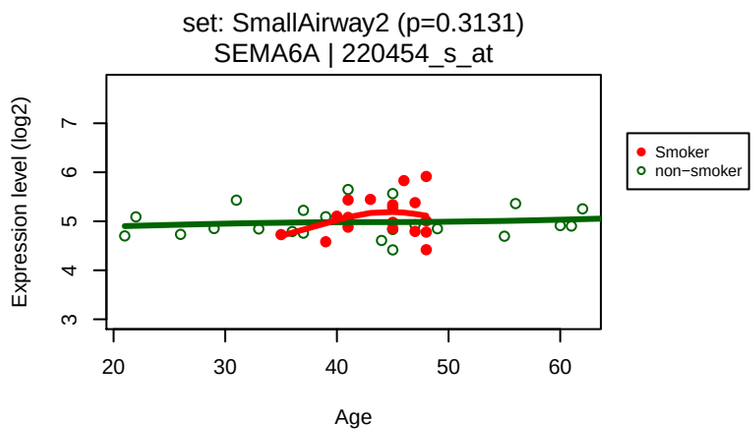
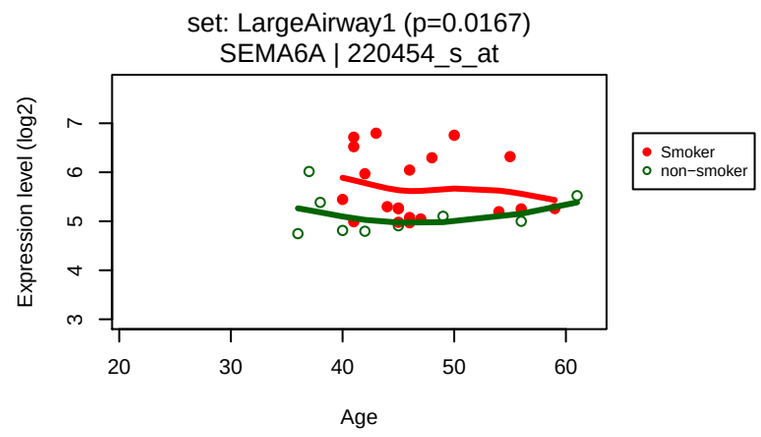

Gene expression of smokers and nonsmokers
gene: SEMA6A | 220454_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1458
p=0.3131
p=0.2698
p=0.0167
p=0.5595
p=0.4671
p=0.4265
p=0.6505
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

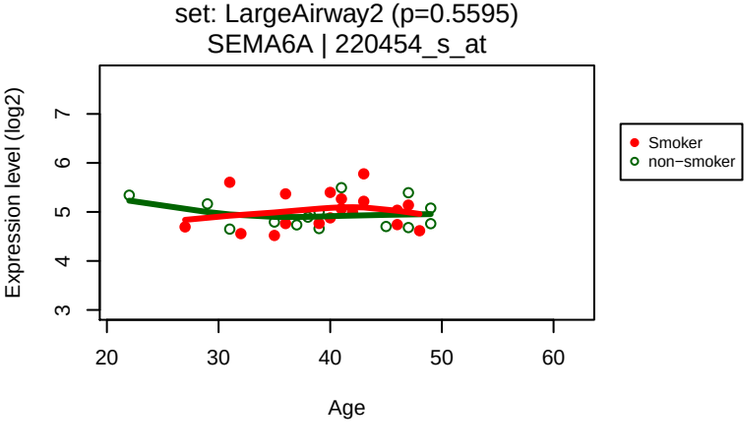

Gene expression ratio (smokers/nonsmokers)
gene: SEMA6A | 220454_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1458
p=0.3131
p=0.2698
p=0.0167
p=0.5595
p=0.4671
p=0.4265
p=0.6505
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

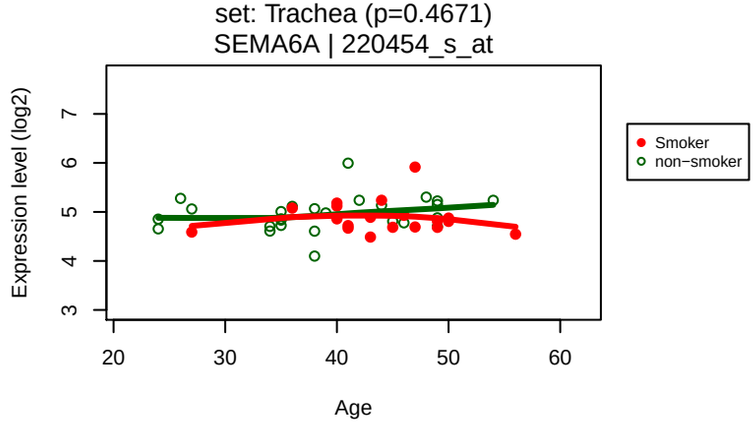

Gene expression ratio in Males
gene: SEMA6A | 220454_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5682
p=0.126
p=0.9868
p=0.9574
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA6A | 220454_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5631
p=0.1916
p=0.3123
p=0.5033
SmallAirway2
LargeAirway2
Trachea
Macrophage2

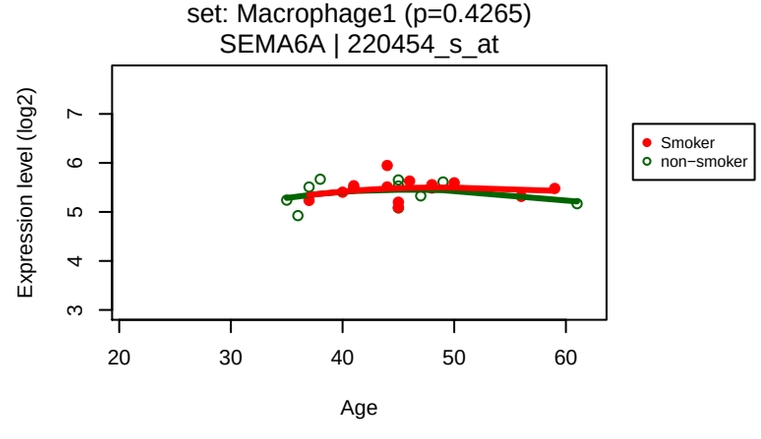

set: SmallAirway1 (p=0.1458)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker

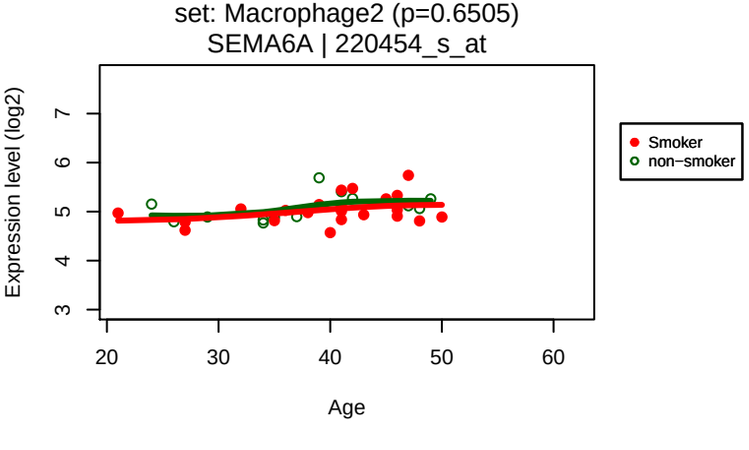

set: SmallAirway2 (p=0.3131)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker

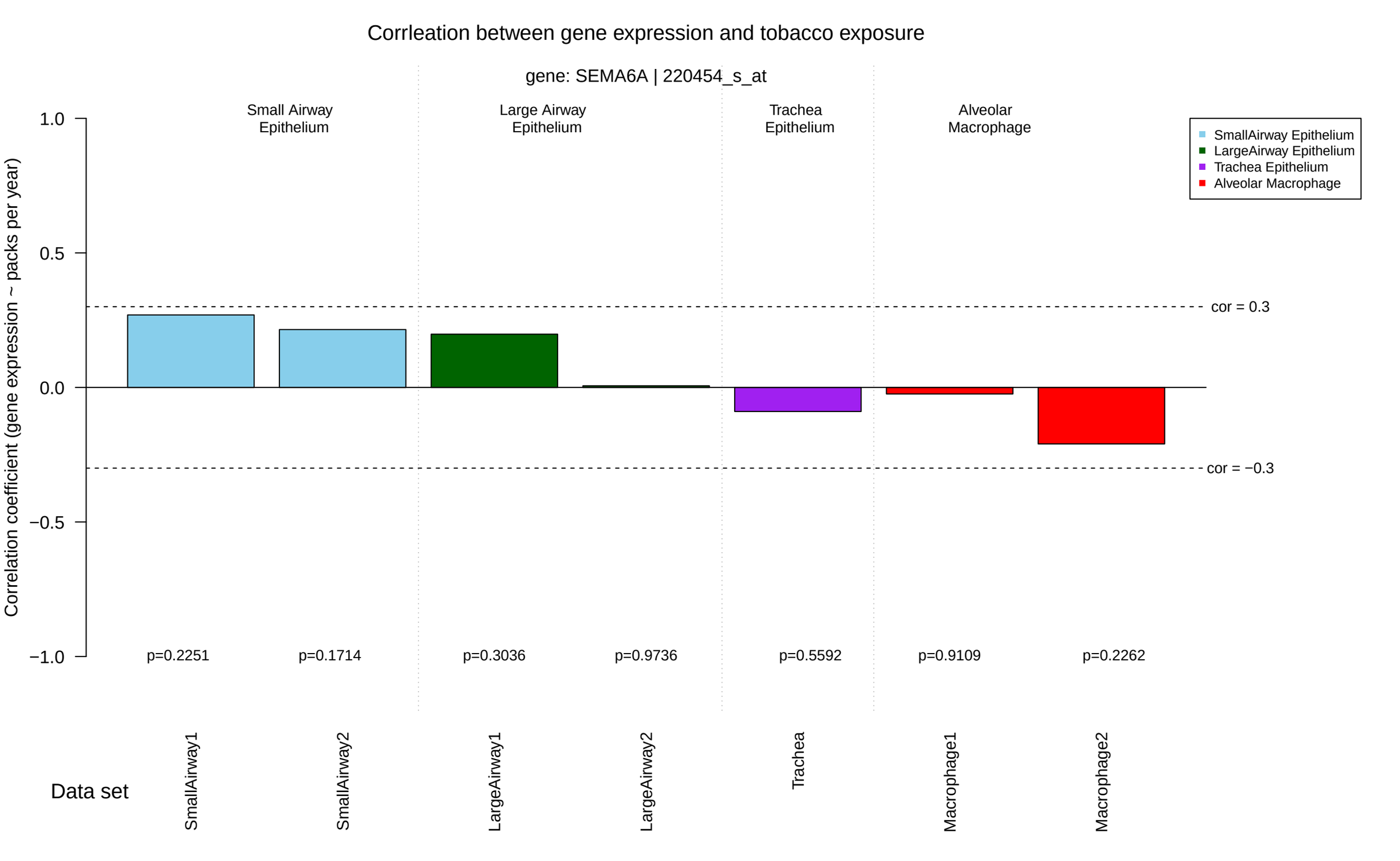

set: LargeAirway1 (p=0.0167)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5595)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4671)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4265)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6505)
SEMA6A | 220454_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA6A | 220454_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2251
p=0.1714
p=0.3036
p=0.9736
p=0.5592
p=0.9109
p=0.2262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2